

Do High Schools Shape College Choices?

Evidence from Centralized Assignment in Chile

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Introduction

High schools may shape consequential choices beyond achievement

- School quality is usually summarized by test scores, but educational production is not one-dimensional. (Jackson 2018; Beuermann et al. 2023; Rose et al. 2022)
- There are important gaps in application and enrollment behavior across groups even when conditioning on achievement. (Hoxby and Turner 2015; Dynarski et al. 2021; Hastings et al. 2016; Chetty et al. 2025)
- Returns differ substantially across institutions and especially across fields. (Kirkeboen et al. 2016)

The paper asks whether schools causally shape college choices

Main question

Do high schools causally shape college choices?

- If so, are those effects fully predicted by achievement value added?
- Do the effects vary with students' baseline achievement?

Answering these questions requires separating school effects from student sorting.

The empirical design proceeds in two steps

- **Estimate outcome-specific value added.** I link Chile's centralized high-school assignment records to enrollment histories, standardized achievement, and higher-education enrollment.
- **Validate it causally.** I estimate school value added in a broad national panel, then test its causal content using random tie-breakers in assignment.

Higher-value-added schools change both achievement and college choices

- Lottery-induced attendance at higher-value-added schools improves math, verbal achievement, entry into high-premium institutions and fields, and projected program income.
- Moving from the 10th to the 90th percentile predicts a **0.435 SD** gain in math and a **0.057 log-point** gain in projected program income.
- Achievement value added does *not* summarize the school dimensions that predict college-field choices and projected income.
- Math effects vary with baseline achievement; the higher-education choice effects do not.

Context and Data

Chile links national school histories to later college choices

- Formal secondary education begins in grade 9 and lasts four years.
- Public and private-subsidized schools participate in centralized assignment; private non-subsidized schools account for about 9.3% of the grade-8 universe.
- Scrambled student identifiers link annual enrollment, grades, attendance, standardized achievement, school preferences, lottery numbers, and offers.

Higher-education applications reveal institution and field choices

- The national admissions route combines standardized tests, school GPA-based scores, and students' ranked program preferences.
- Admission occurs at the institution-program level, with each program choosing the weights placed on score components.
- Administrative records contain exam outcomes, applications, admission scores, and first observed enrollment.
- The linked data therefore capture whether students enroll, where they enroll, and which field they choose.

Chile's centralized assignment system creates national lottery variation

- Families submit ranked school lists to the *Sistema de Admisión Escolar* (SAE), which uses student-proposing Deferred Acceptance.
- Priorities and vacancies determine admission; random tie-breakers resolve excess demand within priority groups.
- Conditional on priorities and simulated assignment risk, tie-breaker offers generate quasi-experimental variation. (Abdulkadiroğlu et al. 2017)
- Replicating the mechanism with the original lottery numbers reproduces more than **99.5%** of first-round assignments in every 2018–2021 cohort.

Linked administrative records connect grade 4 to college enrollment

Sample	Students
Grade-8 cohorts, 2017–2020	944,510
Complete value-added control set	757,999
Timely SAE applicants with assignment risk	226,195
Complete lottery control set	139,317
Lottery sample with math or verbal outcome	95,705

Before high school

Grade-4 achievement, family background, middle-school grades and attendance

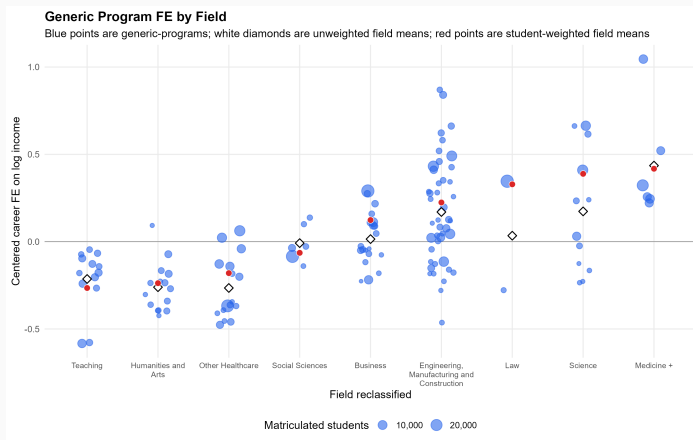
High school

Assignment portfolios, offers, and annual enrollment across 4,133 schools

After high school

Admission-test scores, first college enrollment, institution, field, and projected income

Projected-income premia vary sharply across fields



The high-premium-field outcome includes Science, Law, Engineering/Manufacturing/Construction, and Medicine+.

Three outcomes describe the college option students enter

High-premium field

- Science, Law, Engineering, Manufacturing, Construction, or Medicine+.

High-premium institution

- Institution income premium above 0.1 log points.

Log projected program income

- MiFuturo income predictions from program and institution cells;
- non-enrollment enters at a minimum-wage proxy.

Outcome convention

Non-enrollment is coded as zero for the two premium indicators and enters projected income at a minimum-wage proxy.

Methodology and Empirical Framework

The challenge is sorting across many schools and fallback options

Selection

- Families sort on resources, preferences, location, information, and expectations.
- These same factors predict later college choices.

Many treatments

$$D_i^{\text{full}} = (\mathbf{1}\{S_i = 1\}, \dots, \mathbf{1}\{S_i = J\}).$$

Different counterfactual schools

- An offer to school A can move one student from G^1 and another from G^2 .
- Fully flexible effects become target-school-by-fallback objects.

$$\beta_{AG^1}, \beta_{AG^2}, \dots, \beta_{CG^K}.$$

I project high-school attendance onto outcome-specific scalar value-added indices.

The design estimates and then validates outcome-specific school value added

1. Broad national panel

Estimate observational school value added for each outcome using rich pre-treatment controls.

$$\hat{V}_j^m$$



2. SAE lottery sample

Test whether an offer to a higher-value-added school causally improves the same outcome.

$$\theta^m$$

Does the observational difference encoded in $\hat{V}_j^m$ **pass through** when school attendance changes because of the lottery?

Empirical Framework

Step 1: Estimate observational value added in the broad panel

$$Y_i^m = \alpha^m \lambda_{s(i)}^m + X_i' \Gamma^m + u_i^m, \quad \hat{V}_j^m = \hat{\lambda}_j^m - \bar{\lambda}_\omega^m.$$

- Estimate a separate school fixed effect for math, verbal achievement, high-premium institution, high-premium field, and log projected income.
- X_i includes grade-4 achievement polynomials, family background, middle-school grades and attendance, cohort, gender, age, and home location.
- These measures remain observational: sorting not absorbed by X_i can contaminate $\hat{V}_j^m$.
- The lottery step asks whether the measured school differences contain causal signal.

Empirical Bayes shrinks less precise school estimates more strongly

$$\hat{\tau}_m^2 = \max \left\{ \text{Var}_\omega(\hat{V}_j^m) - \sum_j \omega_j (se_j^m)^2, 0 \right\} \quad \lambda_j^m = \frac{\hat{\tau}_m^2}{\hat{\tau}_m^2 + (se_j^m)^2}, \quad \hat{V}_j^{m,EB} = \lambda_j^m \hat{V}_j^m.$$

- Large, precisely estimated schools remain close to their original fixed effects.
- Small or noisy schools move toward the student-weighted mean of zero. (Walters 2024)
- Shrinkage improves the signal-to-noise ratio of the index used in the lottery design; it does not create a separate causal estimator.

Step 2: Collapse each student's assignment risk to expected value added

Assignment portfolio

$$p_i = (p_{i1}, \dots, p_{iJ}), \quad p_{ij} = \Pr(Z_i = j).$$

Repeated simulations of the assignment mechanism recover each probability.

Outcome-specific risk control

$$E_i^{m,EB} = \sum_j p_{ij} \hat{V}_j^{m,EB}.$$

This is the school value added student i expects over repeated lottery draws, holding the submitted application and priorities fixed.

Identification intuition

The lottery offer's value added, relative to the portfolio's expected value added, is quasi-experimental conditional on assignment risk.

Offered-school value added instruments attended-school value added

Lottery offer

$$O_i^{m,EB} = \hat{V}_{Z_i}^{m,EB}$$

Exogenous conditional on
assignment risk

School attended

$$A_i^{m,EB} = \hat{V}_{D_i}^{m,EB}$$

Endogenous treatment

Student outcome

$$Y_i^m$$

Same outcome used to
define value added

$$Y_i^m = \alpha^m \theta^m A_i^{m,EB} + \rho^m E_i^{m,EB} + W_i' \delta^m + \eta_i^m, \quad A_i^{m,EB} \text{ instrumented by } O_i^{m,EB}.$$

The forecast coefficient measures causal pass-through

$$\theta^m$$

$$\theta^m = 0$$

Observational value added has no causal predictive content along the lottery margin.

$$0 < \theta^m < 1$$

Value added is directionally informative but overstates the lottery-induced causal gain.

$$\theta^m = 1$$

The observational difference passes through one-for-one into the causal outcome difference.

This is a validation design for school value added, following the forecast-coefficient logic of Angrist et al. (2017).

Results

Schools differ meaningfully on achievement and college-choice value added

Outcome	EB VA SD	P10	P90
Math	0.222	-0.195	0.297
Verbal	0.150	-0.172	0.202
High-premium institution	0.106	-0.090	0.144
High-premium field	0.053	-0.060	0.063
Log projected income	0.072	-0.093	0.090

- The 90–10 gap is 0.492 SD for math and 0.374 SD for verbal achievement.
- For projected income, the 90–10 school gap is 0.183 log points.
- The lottery design tests how much of these observational gaps is causal.

Value added predicts causal gains in all five outcomes

	Achievement		Higher-education choices		
	Math	Verbal	High-premium inst.	High-premium field	Log proj. income
θ^{EB}	0.884*** (0.073)	0.885*** (0.087)	0.497*** (0.111)	0.803*** (0.100)	0.311*** (0.103)
N	93,438	95,107	95,651	90,338	95,651

- Achievement pass-through is close to one, despite the absence of immediately lagged test scores.
- School value added also predicts causal movement into higher-premium institutions and fields, and programs with higher projected income.
- I reject $\theta = 0$ in every dimension.

Robust standard errors in parentheses. All models control for expected EB value added and baseline covariates. *** $p < 0.01$.

The estimates imply economically meaningful 90–10 school differences

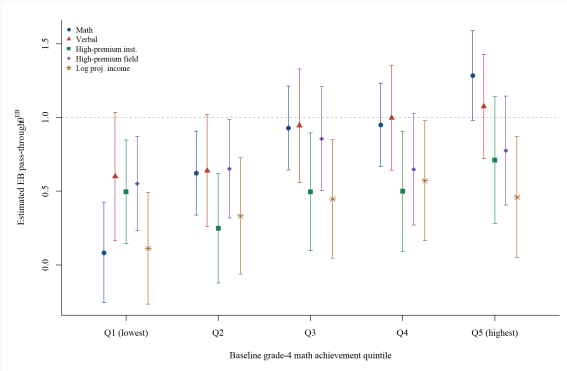
Outcome	P90–P10 EB VA	Implied causal gain
Math	0.492 SD	0.435 SD
Verbal	0.374 SD	0.331 SD
High-premium institution	0.234	11.6 pp
High-premium field	0.123	9.9 pp
Log projected income	0.183 log points	0.057 log points

Interpretation

Along the SAE lottery margin, moving across the observed school value-added distribution changes not only achievement but also the institution, field, and projected earnings of the program students enter.

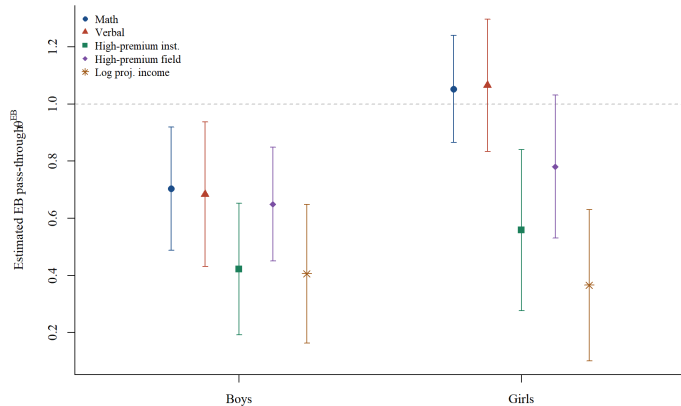
Implied gains equal $\hat{\theta}^{EB} \times (P90 - P10)$ of the corresponding EB value-added distribution.

Math pass-through is weakest for students with low baseline achievement



- Math: $\hat{\theta}_{Q1} = 0.083$ versus $\hat{\theta}_{Q3} = 0.929$.
- Difference: -0.846 ($p < 0.01$).
- The top-quintile difference is positive and marginally significant.
- Higher-education pass-through is stable across baseline math quintiles.

Achievement value added is more predictive for girls



- Math: girls' pass-through is 0.349 higher ($p = 0.017$).
- Verbal: girls' pass-through is 0.382 higher ($p = 0.029$).
- Gender differences are small and statistically insignificant for all three higher-education outcomes.

Field-choice value added is related to, but not reducible to, achievement

Achievement dimensions

$$\text{Corr}(V^{\text{Math}}, V^{\text{Verbal}})$$

0.659

Achievement versus field choice

$$\text{Corr}(V^{\text{Math}}, V^{\text{Field}})$$

0.273

Schools that raise achievement are somewhat more likely to shift students toward high-premium fields, but the two school-value dimensions remain largely distinct.

College-choice value added predicts projected income beyond math

Ordered value-added horse race

1. Math value added enters first.
2. Institution value added enters net of math.
3. Field value added enters net of the first two.

Attended values are jointly instrumented by offered-school values, controlling for each assignment-risk expectation.

School VA dimension	Log proj. income
Math	-0.056 (0.057)
Institution net of math	0.286** (0.143)
Field net of first two	0.366*** (0.096)
<i>N</i>	95,621

Result

Math value added does not carry the projected-income signal once the higher-education choice dimensions are included.

Conclusion

Achievement-based measures of school quality are incomplete

- Outcome-specific observational value added predicts lottery-induced causal gains in achievement, institution choice, field choice, and projected program income.
- Achievement pass-through varies by baseline achievement and gender, while higher-education pass-through is comparatively stable across these groups.
- College-choice value-added dimensions contain projected-income signal beyond math value added, so score-centered accountability can miss consequential school effects on fields and institutions.

**High schools shape not only how much students learn,
but also the options they perceive and choose.**

Sample construction

Sample definition	Students
Linked grade-8 universe, 2017–2021	1,197,982
Cohorts 2017–2020	944,510
+ valid high-school RBD and age 12–16	931,547
+ complete value-added control set	757,999
+ math or verbal admission-test outcome	571,110
Cohorts 2018–2021	959,514
+ timely SAE application	409,737
+ assignment risk	226,195
+ complete lottery control set	139,317
+ math or verbal admission-test outcome	95,705

Different fallbacks make school-specific IV effects unwieldy

- With a common guaranteed fallback G , lottery-induced comparisons could be written as

$$\beta_{AG}, \quad \beta_{BG}, \quad \beta_{CG}.$$

- In practice, students' fallback schools differ:

$$G_i \in \{G^1, G^2, \dots, G^K\}.$$

- The relevant objects become target-school-by-fallback effects:

$$\beta_{AG^1}, \beta_{AG^2}, \dots, \beta_{CG^K}.$$

- The scalar value-added design instead asks whether lottery-induced movement toward schools with higher measured value causes better outcomes.

Formal first stage

$$A_i^{m,EB} = a^m \kappa^m O_i^{m,EB} + \rho_A^m E_i^{m,EB} + W_i' \delta_A^m + \nu_i^m.$$

- $A_i^{m,EB}$ is the EB value added of the attended high school.
- $O_i^{m,EB}$ is the EB value added of the first-round offered school.
- $E_i^{m,EB}$ is expected EB value added under the student's simulated assignment probabilities.
- W_i contains cohort, gender, age, and grade-4 math and language controls.

Ordered value-added IV

$$Y_i^{PI} = \alpha + \theta_M A_i^M + \theta_I A_i^{I \perp M} + \theta_F A_i^{F \perp M, I} \\ + \rho_M E_i^M + \rho_I E_i^{I \perp M} + \rho_F E_i^{F \perp M, I} + W_i' \delta + \eta_i.$$

- A_i^M : math value added.
- $A_i^{I \perp M}$: institution value added residualized on math value added.
- $A_i^{F \perp M, I}$: field value added residualized on the first two dimensions.
- Each attended-school value is instrumented with the corresponding first-round offered-school value.

References: Multidimensional school quality

-  Beuermann, Diether W, C Kirabo Jackson, Laia Navarro-Sola, and Francisco Pardo (Jan. 2023). **“What is a Good School, and Can Parents Tell? Evidence on the Multidimensionality of School Output”**. In: *The Review of Economic Studies* 90.1, pp. 65–101. (Visited on 09/12/2025).
-  Jackson, C. Kirabo (Oct. 2018). **“What Do Test Scores Miss? The Importance of Teacher Effects on Non–Test Score Outcomes”**. In: *Journal of Political Economy* 126.5, pp. 2072–2107. (Visited on 05/31/2026).
-  Kirkeboen, Lars J., Edwin Leuven, and Magne Mogstad (Aug. 2016). **“Field of Study, Earnings, and Self-Selection*”**. In: *The Quarterly Journal of Economics* 131.3, pp. 1057–1111. (Visited on 09/12/2025).
-  Rose, Evan K., Jonathan T. Schellenberg, and Yotam Shem-Tov (July 2022). **The Effects of Teacher Quality on Adult Criminal Justice Contact**. Working Paper. (Visited on 03/30/2026).

References: College choice and information

-  Chetty, Raj, David J Deming, and John N Friedman (Oct. 2025). **“Diversifying Society’s Leaders? The Determinants and Causal Effects of Admission to Highly Selective Private Colleges*”**. In: *The Quarterly Journal of Economics*, qjaf050. (Visited on 10/31/2025).
-  Dynarski, Susan, C. J. Libassi, Katherine Micheltmore, and Stephanie Owen (June 2021). **“Closing the Gap: The Effect of Reducing Complexity and Uncertainty in College Pricing on the Choices of Low-Income Students”**. en. In: *American Economic Review* 111.6, pp. 1721–1756. (Visited on 10/31/2025).
-  Hastings, Justine S., Christopher A. Neilson, Anely Ramirez, and Seth D. Zimmerman (Apr. 2016). **“(Un)informed college and major choice: Evidence from linked survey and administrative data”**. In: *Economics of Education Review*. Access to Higher Education 51, pp. 136–151. (Visited on 09/13/2025).
-  Hoxby, Caroline M. and Sarah Turner (May 2015). **“What High-Achieving Low-Income Students Know about College”**. en. In: *American Economic Review* 105.5, pp. 514–517. (Visited on 10/31/2025).

References: Assignment and methods

-  Abdulkadiroğlu, Atila, Joshua Angrist, Yusuke Narita, and Parag Pathak (2017). **“Research Design Meets Market Design: Using Centralized Assignment for Impact Evaluation”**. In: *Econometrica* 85, pp. 1373–1432. (Visited on 11/02/2025).
-  Angrist, Joshua D., Peter D. Hull, Parag A. Pathak, and Christopher R. Walters (May 2017). **“Leveraging Lotteries for School Value-Added: Testing and Estimation*”**. In: *The Quarterly Journal of Economics* 132.2, pp. 871–919. (Visited on 03/28/2026).
-  Walters, Christopher (Jan. 2024). **“Chapter 3 - Empirical Bayes methods in labor economics”**. In: *Handbook of Labor Economics*. Ed. by Christian Dustmann and Thomas Lemieux. Vol. 5. Elsevier, pp. 183–260. (Visited on 11/03/2025).